

WASP-3b

R-1348

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-3_150810 · created 2026-07-31T22:25:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457244.6855 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.042
Transit depth (Rp/Rs)^2	1.9 +/- 1.15 [%]
Orbital Inclination (inc)	85.04 +/- 2.88
Airmass coefficient 1 (a1)	0.033 +/- 0.018
Airmass coefficient 2 (a2)	3.0 +/- 1.64
Scatter in the residuals of the lightcurve fit is	49.16 %
Best Comparison Star	#3 - [174, 36]
Optimal Aperture	2.64
Optimal Annulus	4.0
Transit Duration (day)	0.0858 +/- 0.0086

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.138 $\pm$ 0.042 vs 0.10538 (deviation 0.7 σ)
Duration fit vs published	0.0858 d vs 0.1167 d (deviation 3.25 σ)
Mid-time O-C	-2.7 min
Depth SNR	3.0
Residual scatter	49.16 %

Light curve

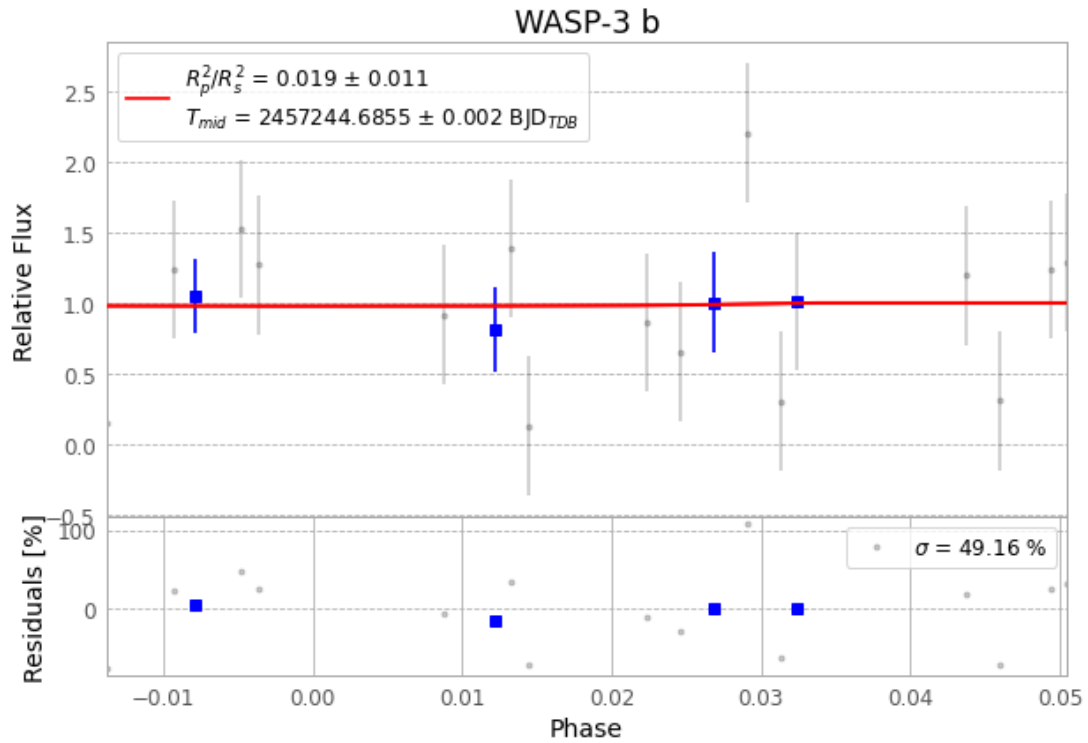


Figure 1 — FinalLightCurve_WASP-3 b_2015-08-09.png (SHA-256 f9da886429628c89...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [42, 386], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e64bd3249a5b89b7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

60 FITS frames (39 MB), WASP-3150810033912.FITS → WASP-3150810063612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-150810.log (350469998b67...), FinalLightCurve_WASP-3 b_2015-08-09.png (f9da88642962...), FinalLightCurve_WASP-3 b_2015-08-09.csv (3274904f4333...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 22:25Z.